

Wireless and RF Attacks

CHAPTER 4

4.01 - Wireless Exploits, Part 1

Episode title: **Wireless Exploits, Part 1**

Objective: **3.2 Given a scenario, research attack vectors and perform wireless attacks**

WIRELESS AND RF VULNERABILITIES

- **Wireless and RF vulnerabilities**

- Broadcast is wide open
- Aircrack-ng

- **Evil twin – rogue WAP used to eavesdrop**

- KARMA attacks (KARMA)
 - Radio machines automatically
 - Denial of service attack

WIRELESS AND RF VULNERABILITIES

- **Deauthentication attacks**

- DoS attacks, disrupt communication between user and WAP

- **Fragmentation attacks**

- DoS attack, floods a network with datagram fragments

WIRELESS AND RF VULNERABILITIES

- **Credential harvesting**

- Process of capturing or discovering valid login
- Social engineering, etc

- **WPS implementation weaknesses**

- Several consumer grade WAPs could allow an attacker to learn the WPS PIN

OTHER WIRELESS VULNERABILITIES

- **Bluejacking**

- Unsolicited messages to a Bluetooth-enabled device

- **Bluesnarfing**

- Stealing information from Bluetooth-enabled device

- **RFID Cloning**

- Unauthorized copy of device's RF signal

OTHER WIRELESS VULNERABILITIES

- **Jamming**

- DoS attack, disables communication among devices

- **Repeating**

- Receiving and retransmitting a signal to increase range

4.02 – Wireless Exploits, Part 2

Episode title: **Wireless Exploits, Part 2**

Objective: **3.2 Given a scenario, research attack vectors and perform wireless attacks**

WIRELESS EXPLOITS

- **Data modification**
 - Unauthorized wireless access to change actionable data
 - May focus on altering configuration to ease further access/attack
- **Data corruption**
 - Attack on availability
 - Targets may be valuable data and/or configuration settings
- **Capture handshakes**
 - Useful in replay and impersonation attacks
- **On-path**
 - Previously known as man-in-the-middle attacks